

Impact of boundary-spanning search on firm innovation performance: a strategic knowledge integration perspective

Xiaoli Li, Zihan Peng and Kun Li

Xiaoli Li, Zihan Peng and Kun Li are all based at the School of Management, Wuhan Textile University, Wuhan, China.

Abstract

Purpose – This study aims to explore the mechanism of boundary-spanning search on firm's innovation performance under environmental dynamics from the perspective of strategic knowledge integration.

Design/methodology/approach – A survey was conducted among Chinese firm managers and R&D personnel, resulting in the collection of 315 valid samples. Hierarchical regression analysis was mainly adopted to demonstrate the hypothesized relationships, while the Sobel test and bootstrap method were used to further validate the mediating effects.

Findings – The results demonstrate that boundary-spanning search in different dimensions is a critical factor in the improvement of firm innovation performance (FIP). Two types of strategic knowledge integration are the main factors causing FIP and mediate the influence of boundary-spanning search on FIP. Furthermore, environmental dynamics moderate the relationship among boundary-spanning search, strategic knowledge integration and FIP.

Practical implications – Managers need to strengthen the boundary-spanning search for market and technical knowledge, which will promote firm innovative performance. Managers also need to implement strategic knowledge integration, which specifically includes using planned strategic knowledge integration to compensate for knowledge deficiencies, thereby achieving predetermined objectives; and using emergent strategic knowledge integration to update their understanding of internal and external environments, and to reset strategic objectives. In dynamic environments, managers should emphasize strategic knowledge management activities more.

Originality/value – From a strategic management perspective, this study categorizes strategic knowledge integration into planned and emergent forms. By applying the logic of knowledge acquisition, integration and creation, it explores how boundary-spanning search affects FIP through strategic knowledge integration as the intermediary and the boundary conditions of environmental dynamics. This not only provides a deeper understanding of the nature and effects of boundary-spanning research but also enhances the theory of strategic knowledge management.

Keywords Boundary-spanning search, Emergent strategic knowledge integration, Environmental dynamics, Planned strategic knowledge integration

Paper type Research paper

1. Introduction

In the era of global economic integration, rapid technological advancements have become a defining feature. Continuous innovation has emerged as a crucial factor for firms to bolster their sustainable competitiveness (Geroski *et al.*, 1993; Lu *et al.*, 2023; Ul-Durar *et al.*, 2023). The knowledge-based view emphasizes that knowledge stands as the most valuable resource and a prerequisite for firms to attain sustained competitive advantage (Asimakopoulous *et al.*, 2020; Yang *et al.*, 2021; Li and Li, 2023). However, in practice, numerous firms often grapple with limited knowledge reserves, posing challenges in meeting innovation demands (Du *et al.*, 2018). Consequently, firms find themselves compelled to seek knowledge but are frequently

Received 7 July 2023
Revised 30 November 2023
22 May 2024
Accepted 24 July 2024

The authors declare that the paper is compliant with ethical standards and that they have no conflicts of interest. The paper is based on the study funded by the National Social Science Foundation of China (grant number 19BGL040).

hindered by their existing organizational and technological environments. This “local search bias” arises from knowledge accumulation grounded in existing knowledge (Cohen and Levinthal, 1990). Moreover, previous research indicates that as organizations mature, their innovation primarily revolves around incremental improvements and operational stability, leading to constraints in exploring new knowledge within their current organizational and technological trajectories (Cohen and Levinthal, 1990; Stuart and Podolny, 1996).

At the same time, as knowledge becomes more dispersed, the gap between the source of innovation and the existing knowledge base continues to widen. To uphold a competitive edge, firms must consistently transcend boundaries and introduce diverse, valuable knowledge (Yu *et al.*, 2019; Wang *et al.*, 2019; Zhu *et al.*, 2024). Within the open innovation model, boundary-spanning search serves as a pivotal method for acquiring heterogeneous knowledge that transcends technological, organizational or geographical boundaries (Phene *et al.*, 2006). Effective boundary-spanning search aids in overcoming core rigidity and capability traps (Levitt and March, 1988; Leonard-Barton, 1992), thereby creating ample opportunities for firm innovation (Zhang and Hu, 2017; Zhao *et al.*, 2018). The significance of boundary-spanning search has been demonstrated in practical applications. For instance, Haier, through its Haier Open Partnership Ecosystem, has facilitated businesses in boundary-spanning knowledge search and integration, leading to the development of over 300 disruptive products (Gao *et al.*, 2021). Baidu has achieved disruptive innovation by acquiring resources through boundary-spanning, establishing an open platform for Apollo autonomous driving and revolutionizing the traditional automobile driving model. Tesla has leveraged external stakeholder resources to establish an OTA car network, disrupting the traditional model (Jin *et al.*, 2024). However, boundary-spanning search needs to consider the types of search boundaries being crossed (Duan *et al.*, 2023), including physical, technological or geographical boundaries, while prior studies often narrowly define boundary-spanning search as organizational search behavior and equate it with external search (Grimpe and Sofka, 2009; Leiponen and Helfat, 2010; Zhu *et al.*, 2022).

Furthermore, innovation can be fueled by either technological progress or market demand (Priem *et al.*, 2012; Zhang *et al.*, 2019; Qu and Mardani, 2023). Consequently, corresponding boundary-spanning search strategies can take the form of market-oriented or technology-oriented search (Sidhu *et al.*, 2004; Zhang *et al.*, 2017), which better reflect the driving effect on innovation. On the other hand, market knowledge search involves collecting information on consumer demand, competitor behavior, supplier characteristics and market trends (Wang *et al.*, 2023). This type of search can reflect a boundary-spanning nature, as firms may need to look for information in markets that differ from their traditional markets or geographic locations to gain new insights and opportunities. Technology knowledge search involves gathering information on new technologies, product development, process flow and R&D dynamics, which can also reveal a boundary-spanning nature, as organizations may need to seek knowledge from other disciplines, technology domains or practices in different regions to promote technological innovation or address specific technological challenges (Wang *et al.*, 2023). However, previous research has most commonly divided boundary-spanning search into depth and breadth of search (Katila and Ahuja, 2002; Wang *et al.*, 2020a; Rui and Lyytinen, 2019), which fails to reflect the type of knowledge that drives innovation or the type of boundary-spanning involved. Innovation is essentially about identifying and using new and distinctive knowledge to drive technological breakthroughs (Van de Ven, 1986; Zhang *et al.*, 2017). Therefore, distinguishing between technological and market knowledge in boundary-spanning search will enhance the focus on these specific types of knowledge search and boundary-spanning activities, enabling innovation behaviors to be more targeted. In addition, some scholars argue that as firms continue to search for external knowledge, it not only leads to knowledge redundancy but also increases search costs, hindering the development of innovation activities (Terjesen and Patel, 2017). It can be seen that previous research on the influence of boundary-spanning search on firm innovation performance (FIP) has yielded different empirical results. Therefore, it is necessary to explore further the relationship between

boundary-spanning search for technological and market knowledge and FIP and clarify the mechanism between the two (Wang *et al.*, 2020b; Chen *et al.*, 2023).

Although boundary-spanning search has brought about a broader range of knowledge, it has also led to the dispersion and fragmentation of knowledge resources (Lemon and Sahota, 2004; Rui and Lyytinen, 2019). Therefore, the extent to which the knowledge from boundary-spanning search can be transformed into an innovation advantage for firms largely hinges on the firm's capability for integrating knowledge (Guo *et al.*, 2019; Yang *et al.*, 2021; Shao *et al.*, 2023). Knowledge integration capability refers to a firm's capacity to decode, reconfigure and apply new knowledge while effectively combining it with existing knowledge (Salunke *et al.*, 2019). On the other hand, as competition becomes increasingly fierce, firms must overcome resource constraints with new strategic logic and actions, meaning that dispersed knowledge must be combined in a meaningful way to create strategic value (Alvarez and Barney, 2007; Zahra, 2015; Guo *et al.*, 2020). Indeed, every strategic decision significantly impacts knowledge, skills and core capabilities (Zack, 1999) and a firm's actual knowledge limits its competitive approach (Laihonen and Mäntylä, 2018). Strategic knowledge management is thus essential for firms to maximize strategic value from knowledge. Organizational strategic objectives can guide knowledge management, and knowledge integration and exploitation help to shape organizational strategy. Strategic knowledge integration involves the process of integrating, transforming and reconstructing knowledge from diverse sources and types according to the strategic needs of the firm to achieve predefined strategic objectives or formulate new ones (Archer-Brown and Kietzmann, 2018; Guo *et al.*, 2020), thereby increasing the utilization of knowledge resources, reducing R&D risks and enhancing innovation performance (Mitchell, 2006). It is thus evident that firms not only need to acquire heterogeneous knowledge through boundary-spanning search but also should integrate this heterogeneous knowledge in line with their strategic development requirements to drive innovation. However, there is currently little research on strategic knowledge integration, with existing studies mainly focusing on overall strategic knowledge management (Hughes and Hodgkinson, 2021; David *et al.*, 2021) or its theoretical aspects (Marques *et al.*, 2016; Bolisani and Bratianu, 2017). Consequently, it is crucial to empirically explore the types of strategic knowledge integration and the mediating mechanism through which strategic knowledge integration affects the relationship between boundary-spanning search and innovation performance.

Simultaneously, with increased economic and environmental variables, environmental dynamics have emerged as an important contextual factor influencing FIP (Sadeghi *et al.*, 2019; Ye *et al.*, 2022). Firms perceive dynamic changes in areas such as technology, markets and demand, which will, in turn, affect their innovative behavior choices when formulating strategic decisions (Wang *et al.*, 2015). However, the impact of environmental dynamics on FIP may even lead to contradictory results (Yuan *et al.*, 2021). On the one hand, environmental dynamics provide more opportunities and higher incentives to promote innovation, but on the other hand, it may pressure the firm by increasing causal ambiguity (Pati *et al.*, 2018). Although environmental dynamics make the environment complicated and volatile, it effectively raises the demand for strategic knowledge integration, consequently enhancing the prospects for identifying innovation. Wang *et al.* (2011) also confirmed that a highly dynamic environment enhances the positive relationship between internal resource integration and organizational performance. However, previous research has not explored the specific circumstances in which strategic knowledge integration plays a role, and therefore, cannot comprehensively reveal the mechanism through which boundary-spanning search affects innovation performance.

Overall, boundary-spanning search is instrumental in accessing diverse and unique resources, thereby stimulating innovation. However, the creation of strategic value from these dispersed resources necessitates strategic knowledge integration. Despite the

significance of this process, previous studies have failed to establish a consensus on the link between boundary-spanning search and innovation performance. Moreover, there is a notable lack of distinction between boundary-spanning search and external search in the literature. Similarly, strategic knowledge integration and its categorization have received insufficient attention. Furthermore, little exploration has been conducted into how boundary-spanning search, in conjunction with strategic knowledge integration, influences innovation performance in dynamic environments. Consequently, this study aims to address the following inquiries:

- RQ1.* How does boundary-spanning search impact firm innovation performance?
- RQ2.* What role does strategic knowledge integration play in the relationship between s boundary-spanning search and firm innovation performance?
- RQ3.* How do environmental dynamics influence the relationship among boundary-spanning search, strategic knowledge integration and firm innovation performance?

To address the above gaps, drawing on a knowledge-based view and strategic knowledge management theory, we investigate the direct effect of boundary-spanning search for technological and market knowledge on FIP, as well as the mediating role of strategic knowledge integration, including planned and emergent strategic knowledge integration. Furthermore, we introduce environmental dynamics as a contingent factor to examine the relationship among boundary-spanning search, strategic knowledge integration and FIP using the moderated mediation model.

We contribute to the literature in three aspects. First, this study enriches the boundary-spanning research literature by clearly defining and categorizing it based on different innovation knowledge types. It also illuminates the pivotal role of boundary-spanning research in driving FIP, thereby providing support for the knowledge-based review. Second, based on the theory of strategic knowledge management, we have categorized strategic knowledge integration into planned and emergent strategic knowledge integration. By using the logic of knowledge acquisition, integration and creation, we have examined the mediating role of strategic knowledge integration in the relationship between boundary-spanning search and FIP, thus enriching the research on strategic knowledge integration. Third, this study expands the contextual mechanism underlying the relationship among boundary-spanning search, strategic knowledge integration and FIP from the perspective of environmental dynamics through a moderated mediation effect model, thereby enhancing our understanding of the intricate interplay among these variables and elucidating the boundary conditions that affect the influence of boundary-spanning search on FIP.

2. Theoretical background and hypotheses

2.1 *Boundary-spanning search and firm innovation performance*

The concept of boundary-spanning search has its origins in the organizational search propositions set forth by [Cyert and March \(1963\)](#). Later, [Nelson and Winter \(1982\)](#) defined the organizational search, highlighting its role in organizations' acquisition of information to identify potential opportunities and generate solutions to challenges. Boundary-spanning search constitutes a crucial component of organizational search ([Wang et al., 2022](#)). Previous research has suggested that boundary-spanning search involves crossing established organizational boundaries to seek and obtain diverse knowledge resources from external entities ([Grimpe and Sofka, 2009](#); [Leiponen and Helfat, 2010](#); [Zhu et al., 2024](#)). However, the boundaries of search vary ([Rosenkopf and Nerkar, 2001](#); [Chesbrough et al., 2018](#); [Zhu et al., 2024](#)), including organizational, technological, geographic and cognitive boundaries. Boundary-spanning search emphasizes the types of search boundaries that should be crossed when acquiring external knowledge ([Duan et al., 2023](#))

rather than merely external search beyond organizational entities. Innovation often requires deviating from existing technological trajectories and acquiring novel and distinctive knowledge to accomplish technological advancements (Van de Ven, 1986; Zhou and Li, 2012). To avoid falling into the capability trap (Levitt and March, 1988), organizations must cross their own physical, technological and geographical boundaries to search for novel and heterogeneous knowledge (Phene *et al.*, 2006). Katila and Ahuja (2002) argue that crossing technological and cognitive boundaries in search activities is essential because it can bring differentiated elements to an organization, increasing knowledge diversity. Based on the theory of open innovation, Ze *et al.* (2018) found that firms can penetrate organizational boundaries and cross cognitive and geographic boundaries to seek out and obtain external innovative knowledge, and then integrate these external knowledge resources with their internal reserves to form the core knowledge assets of the firm. Boundary-spanning search effectively exemplifies the open innovation paradigm.

Rapid technological progress and the tremendous changes in market demand have driven innovation in firms by promoting technological and market knowledge (Nerkar and Roberts, 2004; Qu and Mardani, 2023). If firms have specialized search strategies designed to find technological or market knowledge, they can obtain the highest return on investment (Grimpe and Sofka, 2009). This type of search strategy not only reflects the knowledge sources of innovation effects but also boundaries crossed, such as technology, geography and cognition. The specialized search strategy may be superior to general methods because external knowledge must be transformed to fit into current knowledge reserves (Todorova and Durisin, 2007; Grimpe and Sofka, 2009). To further elaborate on the specialized search strategy, this study divides boundary-spanning search into market and technological knowledge types (Sidhu *et al.*, 2007; Duan *et al.*, 2020). Boundary-spanning search for market knowledge (BSSMK) is the process by which firms actively collect all heterogeneous and valuable knowledge about product usage and substitution patterns, customer preferences and competitors from the market environment (Sidhu *et al.*, 2007; Wei and Atuahene-Gima, 2009). Boundary-spanning search for technological knowledge (BSSTK) includes searches for all heterogeneous and valuable knowledge related to technology and process that affect input–output transformations (Zhang *et al.*, 2019; Wang *et al.*, 2023). Grimpe and Sofka (2009) found that firms primarily acquire technological knowledge from universities, domestic and foreign research institutions, consulting companies and patents, while market knowledge exists mainly in survey organizations, user evaluations and market structures. Dividing boundary-spanning search into technological and market knowledge types is an effective search strategy that can help firms effectively divide market structures and search for potential valuable technologies (Wu *et al.*, 2021).

Based on the knowledge-based view, abundant and unique knowledge resources are considered critical factors for maintaining competitive advantage in organizations (Asimakopoulos *et al.*, 2020; Yang *et al.*, 2021). Boundary-spanning search, which involves seeking external heterogeneous knowledge beyond organizational boundaries, breaks the barriers between products, services, firms, individuals and industries, thereby providing new opportunities for acquiring knowledge resources from diverse sources. This enables firms to obtain innovative and unique knowledge resources, enhancing the diversity of their knowledge base. Through boundary-spanning search, firms can acquire new production processes, technologies, insights from competitors and knowledge of customer preferences. By conducting boundary-spanning searches in distant external sources, firms can identify new ideas and knowledge, overcoming core rigidity and enhancing their innovation capabilities (Leonard-Barton, 1992; Zhang *et al.*, 2022; Hou *et al.*, 2023; Wang *et al.*, 2023). Firms adopt BSSTK to break organizational, technological or geographical constraints and access a more comprehensive array of technological insights. This aids in optimizing production processes, achieving product innovation and developing new products (Lyu *et al.*, 2023). Firms can engage in the BSSTK from various sources, including suppliers, universities and competitors, allowing them to obtain heterogeneous

technological knowledge. This facilitates practical analysis, integration and prediction of technological advancements required for future industry developments (Santoro *et al.*, 2020; An *et al.*, 2022), leading to the launch of new products and high-value-added services and enhancing innovation performance. When conducting a BSSMK, firms engage in search activities related to consumer preferences, supplier dynamics and competitor strategies (Duan *et al.*, 2020). These activities provide firms with a broad and accurate understanding of the product market dynamics and allow them to track customer attitudes and opinions (Gao *et al.*, 2021; Lyu *et al.*, 2022). Through such searches, firms can identify unmet needs in the market and take appropriate measures to lead the development of original products (Lyu *et al.*, 2022). Consequently, acquiring abundant innovative resources through boundary-spanning searches for technological and market knowledge improves innovation performance. Therefore, we hypothesize the following:

- H1a.* Boundary-spanning search for market knowledge has a positive impact on firm innovation performance.
- H1b.* Boundary-spanning search for technological knowledge has a positive impact on firm innovation performance.

2.2 Mediating effect of strategic knowledge integration

The knowledge-based view emphasizes that firms play a central role in knowledge integration, serving as generators and collectors of value rather than relying solely on the market (Grant, 1996; David *et al.*, 2021). Meaningful integration of dispersed knowledge is essential for firms to create strategic value. Boundary-spanning search provides firms with the opportunity to use a variety of strategic knowledge integration techniques, allowing them to identify opportunities for enhancing innovation performance and to engage in a cycle of knowledge acquisition, integration and creation (Lyu *et al.*, 2023). However, it is crucial for knowledge resources to align with the strategic knowledge requirements to bridge the knowledge gap between what the firm needs to know for its strategy execution and its current knowledge base. Mintzberg (1978) classifies strategic formation models into “planned strategy” and “emergent strategy”. Planned strategy involves the deliberate formation of strategies based on formal plans before specific actions, while emergent strategy emphasizes the gradual formation of strategies through trial-and-error learning during operational processes (Slevin and Covin, 1997; Titus *et al.*, 2011). This classification provides a foundation for exploring strategic knowledge integration. Strategic knowledge integration entails synthesizing and using information to achieve predetermined strategic objectives or formulate new strategic goals (Archer-Brown and Kietzmann, 2018; Guo *et al.*, 2020). From the perspective of strategic knowledge management, knowledge integration can be categorized into planned and emergent strategic knowledge integration based on the degree of planning. The former involves actively integrating and transforming specialized knowledge needed by the organization at a strategic level to address knowledge gaps in implementing existing strategies. The latter pertains to the strategic significance of recognizing, sharing and transforming new knowledge to update the organization’s understanding of internal and external environments, thereby formulating new strategies (Guo *et al.*, 2020; David *et al.*, 2021). After strategic integration, the heterogeneous knowledge resources obtained by firms through boundary-spanning search will be reabsorbed, transformed and reconstructed to form a unique knowledge system and exert its value (Blythe and Dadi, 2012; West *et al.*, 2014). This process shapes firm strategy or goal orientation (Bolisani and Bratianu, 2017; David *et al.*, 2021), facilitating the identification of innovation performance opportunities (Guo *et al.*, 2019).

Firms engage in boundary-spanning search to acquire diverse knowledge, enabling them to adjust and develop their strategic knowledge systems, thereby reinforcing their strategic knowledge integration behaviors (Laursen and Salter, 2006). BSSTK primarily focuses on acquiring new knowledge related to processes and technologies, fostering frequent

interactions between firms and universities, research institutions and suppliers, and enhancing the firm's cognitive capabilities in technological knowledge (Huang and Li, 2017). These efforts contribute to the absorption and transformation of knowledge within the firm. Moreover, a BSSTK facilitates the acquisition of advanced technological insights, supporting the establishment of effective knowledge-sharing platforms within the firm and promoting knowledge exchange among internal personnel and external organizations (Men *et al.*, 2019). These activities enable firms to scientifically analyze, integrate and anticipate technological requirements for future industry development (Santoro *et al.*, 2020; An *et al.*, 2022), integrating this knowledge into strategic decision-making and implementing it to reinforce strategic knowledge integration. Through planned strategic knowledge integration, external technical knowledge is strategically combined with internal firm resources, enriching and improving the existing resource combination, aligning technology knowledge with strategic objectives and supporting the organization's established strategic goals. This integrated technical knowledge contributes to long-term technical planning, enhances production processes and improves knowledge utilization efficiency, thereby maintaining a leading position in the technology field and forming a valuable competitive advantage. Furthermore, emergent strategic knowledge integration strengthens understanding of internal and external environments in response to changes in the external market environment and policies. To adapt to new challenges, firms creatively transform technical knowledge into their own business through strategic experiments, learning by doing and formulating new strategic goals, which is beneficial for identifying innovation opportunities (Guo *et al.*, 2020).

Boundary-spanning search encourages firms to interact with multiple parties. In the pursuit of market knowledge through boundary-spanning, firms can strengthen interactions with customers and suppliers, gaining insights into market knowledge from various perspectives such as market feedback, consumer preferences and competitor dynamics, thereby enhancing their connection with the external market (Xu *et al.*, 2017). This not only accelerates the flow of external knowledge but also fosters internal understanding and trust in the process of market interaction, thereby promoting the internalization of knowledge (Liu *et al.*, 2021). Furthermore, it allows for the strategic implications of knowledge absorption from market feedback and customers to be applied to the implementation of strategic objectives (Chandler *et al.*, 2011; Guo *et al.*, 2019), enabling firms to more accurately plan resource allocation, reconsider and reshape their strategic objectives to adapt to changes in their internal and external environments (Vera and Crossan, 2005; Reyman *et al.*, 2015), thus enhancing the efficiency of strategic knowledge integration within firms. Through planned strategic knowledge integration, firms purposefully analyze the direction of market development and continuously promote existing development strategies, facilitating rapid optimization of resource combinations under established strategic objectives and achieving maximum benefits, thereby enhancing innovation performance. Given the rapid changes in the external market, firms require substantial heterogeneous knowledge to support innovation. Only with the ability to perceive market changes in emergent strategic knowledge integration can firms flexibly and comprehensively use newly acquired real-time knowledge from interactions with customers, suppliers, partners and other sources, adjusting their strategies accordingly to identify new opportunities and potential market demands, thereby enhancing innovation performance (Saravathy, 2001; Alvarez and Barney, 2007). Therefore, we propose the following hypotheses:

- H2a.* Planned strategic knowledge integration plays a mediating effect between the boundary-spanning search for market knowledge and firm innovation performance.
- H2b.* Planned strategic knowledge integration plays a mediating effect between the boundary-spanning search for technological knowledge and firm innovation performance.

- H2c.* Emergent strategic knowledge integration plays a mediating effect between the boundary-spanning search for market knowledge and firm innovation performance.
- H2d.* Emergent strategic knowledge integration plays a mediating effect between the boundary-spanning search for technological knowledge and firm innovation performance.

2.3 Moderating effect of environmental dynamics

According to the contingency theory, organizational performance is determined not just by the strategy itself but by the fit between the strategy and the business environment (Donaldson, 2001). Environmental factors have become important contextual factors influencing firm innovation (Sadeghi *et al.*, 2019; Ye *et al.*, 2022). The dynamism of the environment is often caused by intense competition and unpredictable technological advancements (Wang *et al.*, 2015). The perceived environmental dynamism by firms influences their choice of innovative behaviors when making strategic decisions (Kumar *et al.*, 2022). In high-level dynamic environments, there is an increased demand for heterogeneous knowledge (Zhu *et al.*, 2023), which increases the requirement for strategic knowledge integration. Effective knowledge integration facilitates strategic decision-making, identification of market opportunities and formulation of practical responses, thereby enhancing innovation. Therefore, in high-level dynamic environments, the improvement of innovation performance is more likely influenced by the strength of strategic knowledge integration. The stronger the environmental dynamics, the more essential it becomes for firms to timely and accurately introduce new knowledge resources into their organizations, engage in strategic knowledge integration and reconfiguration and integrate knowledge in a planned and timely manner, thus enhancing innovation performance (Sarasvathy, 2001; Reyman *et al.*, 2015). Specifically, in high-level dynamic environments, particularly when facing intense market competition, firms strengthen planned strategic knowledge integration. To achieve the highest expected returns, firms implement strategies through rational analysis and prediction of the environment, integrating complementary knowledge (Bolisani and Bratianu, 2017). This enables firms to effectively identify the deficiencies of existing innovation models and the emerging needs of niche market users (Alvarez and Barney, 2005), helping actively identify points of differentiation for products or services and create new value propositions. At the same time, in high-level dynamic environments, especially during unexpected events, firms strengthen emergent strategic knowledge integration. In response to challenges, firms swiftly and flexibly allocate innovative resources and adjust strategic objectives. Integrating and reconfiguring knowledge resources help firms explore new opportunities, reduce inertial thinking and path dependency (Brettel *et al.*, 2012), and thus respond flexibly to unexpected situations, gaining faster growth opportunities. Therefore, we propose the following hypotheses:

- H3a.* Environmental dynamics positively moderate the relationship between planned strategic knowledge integration and firm innovation performance.
- H3b.* Environmental dynamics positively moderate the relationship between emergent strategic knowledge integration and firm innovation performance.

2.4 Moderated mediating effect

The mediating effect of *H2* and the moderating effect of *H3* synergistically form a moderated mediating effect, where the mediating effect of strategic knowledge integration is moderated by environmental dynamics. In other words, the intensity of planned and emergent strategic knowledge integration will vary depending on the level of environmental dynamics. In high-dynamic environments, firms face more significant market competition pressure due to rapid changes in environments, resulting in their knowledge resources quickly becoming outdated and challenging to adapt. Therefore, in such environments, it

becomes increasingly important for firms to use boundary-spanning search to access diverse knowledge resources for strategic knowledge integration (Zhu *et al.*, 2023). By integrating and reconfiguring these knowledge resources based on their strategic planning, firms can develop new products, expand market share and enhance innovation performance more effectively. In low-dynamic environments, the indirect effect of boundary-spanning search on innovation performance through strategic knowledge integration may weaken. Therefore, we hypothesize the following:

- H4a. Environmental dynamics positively moderates the mediating effect of boundary-spanning search for market knowledge affecting firm innovation performance through planned strategic knowledge integration.
- H4b. Environmental dynamics positively moderates the mediating effect of boundary-spanning search for technological knowledge affecting firm innovation performance through planned strategic knowledge integration.
- H4c. Environmental dynamics positively moderates the mediating effect of boundary-spanning search for market knowledge affecting firm innovation performance through emergent strategic knowledge integration.
- H4d. Environmental dynamics positively moderates the mediating effect of boundary-spanning search for technological knowledge affecting firm innovation performance through emergent strategic knowledge integration.

Based on the above, this study constructs the research model shown in Figure 1.

3. Methodology

3.1 Questionnaire design and data collection

A large-sample survey was conducted to collect data for this study. The research questions of this study revolve around innovative behaviors, so firms with independent innovation capabilities were specifically selected as the research subjects. The samples were primarily drawn from emerging industries such as the internet, biomedicine, electronic equipment manufacturing, communication equipment and new energy vehicles. These samples were diverse and covered various regions in China such as Jiangsu, Hubei, Hunan, Shanghai, Beijing and Sichuan. To ensure questionnaire accuracy and validity, specific criteria were applied when selecting participants. Middle and senior managers who had worked for the firm for at least two years and possessed a good understanding of R&D activities were included. Additionally, R&D personnel who had implemented technology or product innovations in the past three years were also included. There were two main parts to the

Figure 1 Theoretical model

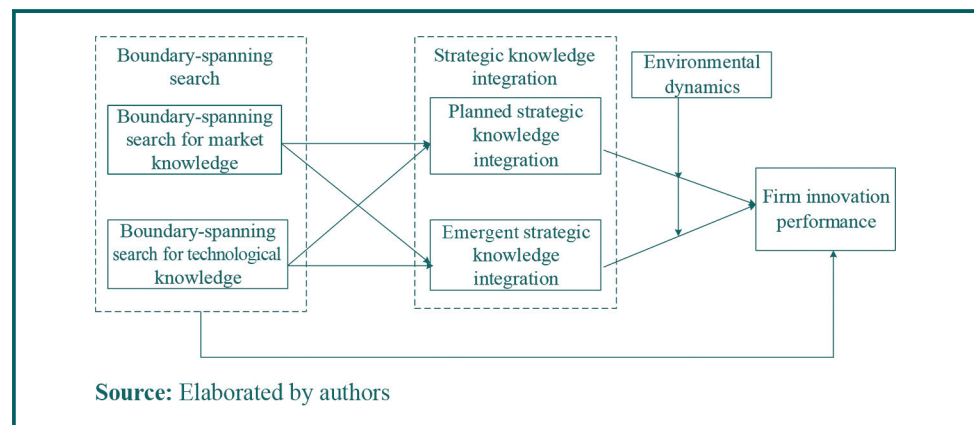


Table 1 Demographic details of the samples ($N = 315$)

Demographics	Category	Count	%	Demographics	Category	Count	%
Gender	Male	101	32.1	Firm size (No. of employees)	0–100	85	26.9
	Female	214	67.9		101–300	67	21.3
Education	College and below	34	10.8		301–1,000	22	7.0
	Bachelor	109	34.6	Firm type	>1,000	141	44.8
	Master and above	172	54.6		State-owned	120	38.1
Tenure (years)	0–1	116	36.8		Private	125	39.7
	2–3	102	32.4	Firm age (years)	Other	70	22.2
	4–5	38	12.1		0–5	63	20.0
	>5	59	18.7		6–10	82	26.0
					>10	170	54.0

Source: Elaborated by authors

questionnaire. The first part collected basic information about the firm, while the second part comprised subjective questions related to boundary-spanning search, strategic knowledge integration, environmental dynamics and FIP. The survey was conducted from December 2020 to June 2021 and then from November 2021 to February 2022. A total of 610 questionnaires were distributed through field visits, WeChat and email. Out of these, 386 completed questionnaires were collected. After excluding 71 questionnaires with missing or apparent errors, 315 final valid questionnaires were obtained, resulting in an effective response rate of 51.64%. The sample characteristics statistics of the questionnaire show a wide range of respondents' working years, business sizes and firm years, indicating a good representation of the sample chosen for this study. Table 1 presents the descriptive statistical analysis results of the sample's characteristics.

3.2 Measurements

We adhere to the standard protocol for developing our survey measurement tools. First, the items for each construct in this study were derived from previous research with minor modifications to suit the specific context of this study. All items for the constructs were scored using a five-point Likert scale, with "1" being "strongly disagree" and "5" being "strongly agree". The measurement items used in this study are detailed in Appendix Table A1. Furthermore, as previously mentioned, given that the survey was carried out in China, the back-translation method was used to ensure consistency between the instrument's Chinese and original English versions (Singh, 1995). This study categorized boundary-spanning search into two constructs: BSSTK and BSSMK. The measurement of these two constructs drew upon scales developed by Sidhu *et al.* (2007), Zhang *et al.* (2019) and Duan *et al.* (2020), with each comprising seven items. Furthermore, strategic knowledge integration was bifurcated into planned strategic knowledge integration (PKI) and emergent strategic knowledge integration (EKI). The measurement of these two constructs was based on studies by Archer-Brown and Kietzmann (2018) and Marques *et al.* (2016), further supported by Guo *et al.* (2020). Both planned and emergent strategic knowledge integration included eight measurement items. The measurement scale for environmental dynamics (ED) was derived from Schilke (2014) and incorporated into the background of this study, using four measurement items. The measurement of FIP was based on existing literature (Li *et al.*, 2008; Forés and Camisón, 2016; Cheng and Shiu, 2015). This construct included six items, encompassing technological innovation aspects, including product innovation, technological innovation and process innovation, as well as incremental and radical approaches to innovation.

In line with prior research, this study includes firm size, firm age and firm type as control variables (Yang *et al.*, 2021; Lyu *et al.*, 2023; Wang *et al.*, 2023). While these control

variables can partially reflect the advantages in resource acquisition or experience level of firms, studies suggest that they may not decisively impact research outcomes (Romero and Martínez-Román, 2012). Larger firms may encounter inflexibility in responding to market changes (Jansen *et al.*, 2005; McIver and Lepisto, 2017), whereas mature firms could be constrained in innovation due to inherent inertia. Furthermore, different types of firms demonstrate variations in resource acquisition advantages; for instance, state-owned firms typically have access to more government funding but may also face unique challenges when pursuing non-profit objectives (Guan *et al.*, 2009; Wang *et al.*, 2021). Consistent with the measurement methods used in prior research (Yang *et al.*, 2021; Lyu *et al.*, 2023; Wang *et al.*, 2023), this study measures firm size using the number of employees, while the number of years since establishment is used to assess firm age. Additionally, the study distinguishes between four types of firms, including state-owned and private firms.

4. Data analysis and research results

4.1 Reliability and validity tests

The reliability of the measured question items in this study was tested using Cronbach's alpha. The results are presented in Table 2, where all six variables exhibited Cronbach's alpha values greater than 0.7, indicating good internal consistency reliability of the scales used in this study. Because the scales used were primarily established measures, they possess strong content validity. To further assess the convergent and discriminant validity of the variables, confirmatory factor analysis (CFA) was conducted and the average variance extracted (AVE) and the composite reliability (CR) were calculated. The results are also presented in Table 2. First, the overall fit indicators of the measurement model were assessed, and the results indicated a good goodness of fit with CMIN/DF = 1.654, RMSEA = 0.046, NFI = 0.903, IFI = 0.959 and CFI = 0.959. Second, the AVE and CR values both surpassed the recommended thresholds of 0.5 and 0.7, respectively, demonstrating strong convergent validity of the scale. Additionally, the square root of AVE was computed for each variable to examine their discriminant validity, as shown in Table 4. The correlation coefficients between the variables were observed to be lower than the square root of the corresponding AVE values, affirming the good discriminant validity of the scale in this study.

4.2 Nonresponse bias and common method bias tests

Before conducting the analysis, we assessed for potential nonresponse bias and common method bias (CMB). When the opinions expressed by nonparticipants significantly differ from those of actual participants, nonresponse bias occurs (Collis and Hussey, 2013). In such cases, the sample may not accurately represent the entire population, thereby limiting the generalizability of the results (Albawwat, 2022). To ensure the sample's representativeness of the population, nonresponse bias tests are conducted. We evaluated nonresponse bias in two ways. First, inspired by Dubey *et al.* (2021), we selected 41 questionnaires that were not submitted within the specified timeframe. Using objective demographic indicators, we compared the data from these questionnaires with the data

Table 2 Measurement model

Variable	No. of items	Cronbach's α	AVE	CR
BSSTK	7	0.921	0.633	0.923
BSSMK	7	0.928	0.660	0.931
PKI	4	0.927	0.786	0.936
EKI	4	0.884	0.661	0.886
ED	4	0.836	0.568	0.839
FIP	6	0.8942	0.596	0.898

Source: Elaborated by authors

from 60 questionnaires submitted within the specified timeframe. The results of this non-response bias test are revealed in Table 3. As demonstrated in Table 3, the results of the independent samples *t*-test indicate no statistically significant differences between the two groups of respondents in terms of respondent characteristics. Second, following the time trend extrapolation method recommended by Armstrong and Overton (1977), we compared the data collected in two stages (157 and 158 responses). The *t*-test analysis yielded non-significant differences. Hence, there appears to be no evident issue of nonresponse bias in this study.

Due to the possibility of CMB in self-reported data (Podsakoff and Organ, 1986; Podsakoff et al., 2003), we adopted rigorous procedures to minimize CMB effects. During the instrument design phase, we ensured the clarity, precision and conciseness of the items and adopted a cross-ordering approach for the items. During the survey distribution phase, we guaranteed the anonymity of the respondents and promised the confidentiality of their answers (Chang et al., 2010; Rindfleisch et al., 2008). After the data collection, we further used multiple methods to test for CMB in our data set. First, we undertook Harman's single-factor test to examine CMB by conducting an exploratory factor analysis (EFA) on key variables involved in the hypothesis tests. The EFA revealed that the first factor accounted for 37.78% of the variance, which is below the threshold of 50%, indicating that CMB is unlikely to be a serious concern in this study (Podsakoff and Organ, 1986). Second, we also used CFA to assess Harman's single-factor test, which is more elaborate (Podsakoff et al., 2003). The results indicated that the one-factor model did not fit the data well ($\chi^2/df = 7.536$, NFI = 0.545, IFI = 0.580, CFI = 0.578, RMSEA = 0.144). Third, we used an unmeasured latent common method factor (ULCMF) to test the two-factor model (Williams and Anderson, 1994; Williams et al., 1996). The comparison between the ULCMF model with a common method factor and the original measurement model revealed no significant differences between the two models, where $\Delta NFI = 0.001$ and $\Delta IFI = 0.001$ are both below the threshold of 0.1. Additionally, Table 4 displays the correlation coefficients among the variables, and the results do not show any highly correlated variables, whereas CMB typically leads to extremely high correlations ($r > 0.90$) (Bagozzi et al., 1991). In summary, these tests indicate that CMB is not a serious issue in our study.

4.3 Descriptive statistics

The means, standard deviations and correlations among the variables are presented in Table 4. The significant correlations ($p < 0.01$) among the variables support the initial hypotheses of this study and form a crucial foundation for subsequent empirical tests.

Table 3 Nonresponse bias tests

Respondent characteristics	Cutoff	N	Mean	Levene's test F	T-test Sig.	t	Sig.
Gender	Beyond	41	1.68	0.480	0.490	0.341	0.734
	Within	60	1.65				
Education	Beyond	41	2.46	0.422	0.518	1.246	0.216
	Within	60	2.28				
Tenure	Beyond	41	2.15	0.286	0.594	0.060	0.952
	Within	60	2.13				
Firm size	Beyond	41	2.66	0.293	0.590	-1.137	0.258
	Within	60	2.97				
Firm type	Beyond	41	2.10	0.296	0.588	0.519	0.605
	Within	60	1.98				
Firm age	Beyond	41	2.32	0.675	0.413	0.303	0.762
	Within	60	2.26				

Source: Elaborated by authors

Table 4 Descriptive statistics and correlations

Variable	Mean	SD	1	2	3	4	5	6	7	8	9
1. Firm size	2.700	1.285	1								
2. Firm type	2.060	1.127	−0.149**	1							
3. Firm age	2.340	0.791	0.415**	−0.317**	1						
4. BSSMK	3.870	1.078	0.069	0.058	−0.112*	0.796					
5. BSSTK	3.133	0.953	−0.014	0.001	−0.059	0.414**	0.813				
6. PKI	3.187	1.280	−0.071	0.108	−0.159**	0.529**	0.468**	0.813			
7. EKI	3.426	1.129	0.077	0.065	−0.025	0.469**	0.541**	0.533**	0.886		
8. ED	3.314	0.887	0.058	0.047	0.097	0.346**	0.374**	0.313**	0.319**	0.754	
9. FIP	3.141	0.879	0.037	0.013	0.001	0.550**	0.600**	0.586**	0.576**	0.593**	0.772

Notes: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; the italicized values on the diagonal are the square root of AVE

Source: Elaborated by authors

Further analysis, including regression analysis, the Sobel test and the bootstrap method, will be used to investigate the interactions between the variables.

4.4 Hypotheses tests

Regression analysis was conducted based on the correlation analysis, and the results are presented in Table 5. After controlling for firm size, nature and duration, the effects of BSSMK (M1, $\beta = 0.458$, $p < 0.001$) and BSSTK (M2, $\beta = 0.556$, $p < 0.001$) on FIP were found to be significantly positive. These findings confirm *H1a* and *H1b*.

To test the mediating effect, this study first adopted the causal steps method (Baron and Kenny, 1986). As shown in Table 5, both BSSMK and BSSTK had a positive influence on planned strategic knowledge integration (M3, $\beta = 0.624$, $p < 0.001$; M4, $\beta = 0.620$, $p < 0.001$), as well as on emergent strategic knowledge integration (M5, $\beta = 0.488$, $p < 0.001$; M6, $\beta = 0.641$, $p < 0.001$). Next, we introduced both BSSMK and planned strategic knowledge integration into the regression model M7, which showed significant effects on FIP ($\beta = 0.275$, $p < 0.001$; $\beta = 0.292$, $p < 0.001$). The regression coefficient of the BSSMK decreased from 0.458 to 0.275 compared to M1, indicating that planned strategic knowledge integration partially mediates the relationship between BSSMK and FIP, confirming *H2a*. In M8, both BSSTK and planned strategic knowledge integration significantly impacted FIP ($\beta = 0.383$, $p < 0.001$; $\beta = 0.279$, $p < 0.001$), with the regression coefficient of BSSTK decreasing from 0.556 to 0.383 compared to M2. This result shows that

Table 5 Results of hierarchical regression analysis of main effects and mediating effects

Variable	FIP		PKI		EKI		FIP		M9	M10	M11
	M1	M2	M3	M4	M5	M6	M7	M8			
Firm size	−0.023	0.025	−0.078	−0.011	0.037	0.088	−0.001	0.028	−0.05	0.001	−0.010
Firm type	0.000	0.021	0.056	0.082	0.053	0.076	−0.016	−0.002	−0.02	0.000	−0.010
Firm age	0.087	0.034	−0.083	−0.168	−0.037	−0.015	0.111*	0.080	0.07	0.038	0.970
BSSMK	0.458***		0.624***		0.488***		0.275***		0.302***		0.181***
BSSTK		0.556***		0.620***		0.641***		0.383***		0.379***	0.276***
PKI							0.292***	0.279***			0.169***
EKI									0.318***	0.276***	0.142***
R^2	0.308	0.363	0.297	0.241	0.224	0.305	0.435	0.489	0.438	0.450	0.742
Adj. R^2	0.299	0.355	0.288	0.231	0.214	0.296	0.426	0.481	0.429	0.440	0.541
F	34.479***	44.240***	32.687***	24.576***	22.411***	34.081***	47.650***	59.157***	48.12***	50.742***	72.579***

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Source: Elaborated by authors

planned strategic knowledge integration partially mediates the relationship between BSSTK and FIP, confirming *H2b*.

We then incorporated both BSSMK and emergent strategic knowledge integration into the model M9. The results showed that both variables significantly impacted FIP ($\beta = 0.302$, $p < 0.001$; $\beta = 0.318$, $p < 0.001$), with the coefficient of BSSMK decreasing from 0.458 to 0.302 compared to M1. Therefore, emergent strategic knowledge integration partially mediates between BSSMK and FIP, confirming *H2c*. In M10, both BSSTK and emergent strategic knowledge integration showed significant effects on FIP ($\beta = 0.379$, $p < 0.001$; $\beta = 0.276$, $p < 0.001$), with the regression coefficient of BSSTK decreasing from 0.556 to 0.379 compared to M2. This result confirmed *H2d*, showing that emergent strategic knowledge integration partially mediates between BSSTK and FIP.

Additionally, we constructed a full model M11, which included the joint effects of boundary-spanning search and strategic knowledge integration on FIP, controlling for firm size, nature and duration. The results revealed that both variables significantly influenced FIP, further supporting *H1* and *H2* in our study.

However, some scholars argue that [Baron and Kenny's method \(1986\)](#) may not fully capture mediating effects ([Preacher and Hayes, 2008](#)). To address this concern, we conducted a Sobel test, which is often used to complement the causal steps approach and empirically evaluate mediating effects. The results in [Table 6](#) show that planned and emergent knowledge integration mediate between boundary-spanning search and FIP, providing strong support for our *H2*. Notably, the Sobel test assumes a normal distribution of ab , which is rarely observed ([Hayes, 2009](#)). To overcome this limitation, we adopted the bootstrap method proposed by [Preacher and Hayes \(2008\)](#), which involves non-parametric resampling of the data set for repeated estimation. We implemented Hayes's bootstrap method with 5,000 bias-corrected sampling times and a 95% confidence interval, and the results of the bootstrap mediation effects are presented in [Table 7](#). As shown in [Table 7](#), when planned and emergent knowledge integration are treated as mediating variables, the confidence intervals do not include zero, thus confirming *H2a*, *H2b*, *H2c* and *H2d* once again. Using the Sobel test and the bootstrap method, we provide more robust evidence for the mediating effects of planned strategic knowledge integration and emergent strategic knowledge integration on the relationship between boundary-spanning search and FIP.

The results of the analysis on moderating effects are presented in [Table 8](#). In this study, the independent and moderating variables are both centrally incorporated into the model. The findings indicate that FIP is significantly influenced by the interaction between

Table 6 Sobel test results of the mediating effect

Mediation path	Test type	Test statistic	Std. Error	p-value
BSSMK → PKI → FIP	Sobel test	6.491881	0.027194	0.000
	Aroian test	6.474236	0.027268	0.000
	Goodman test	6.509671	0.027120	0.000
BSSTK → PKI → FIP	Sobel test	6.280604	0.026854	0.000
	Aroian test	6.260995	0.026938	0.000
	Goodman test	6.300399	0.026770	0.000
BSSMK → EKI → FIP	Sobel test	6.273262	0.024813	0.000
	Aroian test	6.253655	0.024891	0.000
	Goodman test	6.293055	0.024735	0.000
BSSTK → EKI → FIP	Sobel test	6.036516	0.029342	0.000
	Aroian test	6.019799	0.029423	0.000
	Goodman test	6.053374	0.029260	0.000

Source: Elaborated by authors

Table 7 Bootstrap test results of the mediating effect

Mediation path	Effect type	Effect	BootSE	95% Confidence interval	
				BootLLCI	BootULCI
BSSMK → PKI → FIP	Direct effect	0.275	0.037	0.199	0.34
	Mediating effect	0.182	0.026	0.129	0.227
BSSTK → PKI → FIP	Direct effect	0.383	0.041	0.306	0.466
	Mediating effect	0.173	0.024	0.121	0.216
BSSMK → EKI → FIP	Direct effect	0.302	0.036	0.233	0.371
	Mediating effect	0.155	0.024	0.112	0.205
BSSTK → EKI → FIP	Direct effect	0.379	0.037	0.288	0.474
	Mediating effect	0.176	0.031	0.139	0.262

Source: Elaborated by authors

environmental dynamics and planned strategic knowledge integration ($\beta = 0.147$, $p < 0.001$). Furthermore, the introduction of the interaction term increases the R^2 value, indicating an improvement in the model's explanatory power. Similarly, the performance of the firm's innovation process is significantly impacted by the interaction between environmental dynamics and emergent strategic knowledge integration ($\beta = 0.091$, $p < 0.05$), and the inclusion of the interaction term also leads to an increase in R^2 . These results offer empirical support for *H3a* and *H3b*, confirming the hypothesized moderating effects.

Figure 2 illustrates the moderating effects of environmental dynamics at both high ($M + 1SD$) and low ($M - 1SD$) levels. The results from both graphs show that planned and emergent strategic knowledge integration have positive effects on innovation performance across different levels of environmental dynamics. Specifically, high environmental dynamics strengthen the positive link between planned strategic knowledge integration and FIP, as well as the positive relationship between emergent strategic knowledge integration and FIP. Conversely, low environmental dynamics weaken the positive correlation between planned strategic knowledge integration and FIP, and also weaken the positive relationship between emergent strategic knowledge integration and FIP.

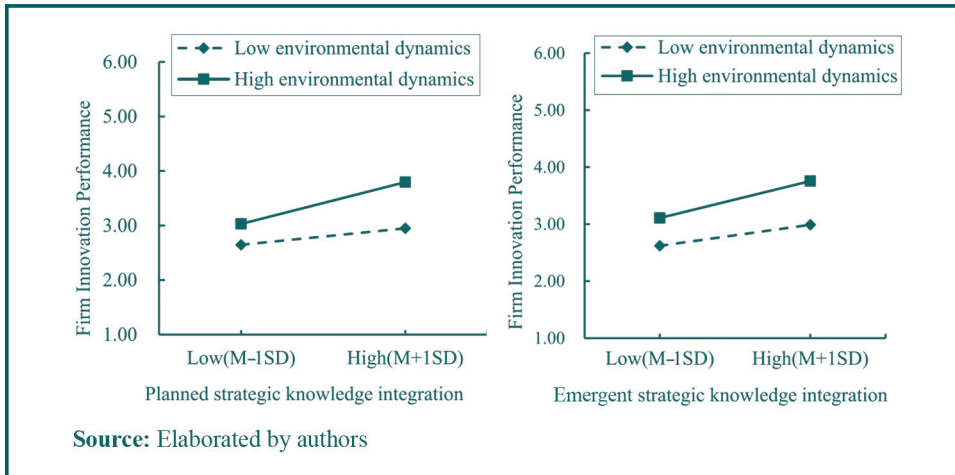
The bootstrap method was used in this study to conduct a moderated mediation effects test. The environmental dynamics were classified into two categories using the mean ± 1 standard deviation as the criteria. The results of the moderated mediation effect test, demonstrated in Table 9, examine the positive moderating role of environmental dynamics in the relationship between boundary-spanning search and FIP through strategic knowledge integration. The findings in Table 9 reveal that the confidence intervals for environmental dynamics at both high and low grouping levels do not include zero. This

Table 8 Analysis of the moderating effect of environmental dynamics

Variable	FIP			
	M1	M2	M3	M4
Firm size	0.025	0.024	−0.008	−0.011
Firm type	−0.041	−0.032	−0.041	−0.039
Firm age	−0.003	0.017	−0.050	−0.046
PKI	0.311***	0.333		
EKI			0.336***	0.356***
ED	0.448***	0.425***	0.459***	0.444***
ED*PKI		0.147***		
ED*EKI				0.091*
R^2	0.534	0.564	0.522	0.531
Adj. R^2	0.527	0.556	0.514	0.522
F	70.884**	66.418**	67.533**	58.199**

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Source: Elaborated by authors

Figure 2 Moderating effect of environmental dynamics

indicates that the conditional indirect effects of boundary-spanning search on FIP through strategic knowledge integration are statistically significant. Furthermore, the corresponding INDEX indicators also do not include zero, indicating that the model of moderated mediation effect is significant. This suggests that *H4a*, *H4b*, *H4c* and *H4d* are confirmed.

5. Discussions and implications

The knowledge-based view emphasizes that resources, particularly diverse knowledge resources, constitute the foundational elements for a firm's competitive advantage. Firms are required to cross various boundaries in pursuit of external knowledge (Sidhu *et al.*, 2007; Segarra-Ciprés and Bou-Llusar, 2018). Despite yielding a broader spectrum of knowledge, boundary-spanning search results in dispersed and fragmented knowledge resources (Lemon and Sahota, 2004). Only through meaningful combination of dispersed knowledge can strategic value be created (Alvarez and Barney, 2007; Zahra, 2015; Guo *et al.*, 2020). Echoing the existing literature, this study investigates how boundary-spanning search effectively governs FIP through strategic knowledge integration as an intermediary

Table 9 A moderated mediation effect

Mediation path	Test	Group	Effect	BootSE	95% Confidence interval	
					BootLLCI	BootULCI
BSSMK → PKI → FIP	Conditional indirect effects	M-1SD	0.0849	0.0394	0.0078	0.1644
		M + 1SD	0.2439	0.0504	0.1479	0.3488
		INDEX	0.082	0.0287	0.0255	0.138
BSSTK → PKI → FIP	Conditional indirect effects	M-1SD	0.0763	0.0263	0.0271	0.1306
		M + 1SD	0.1736	0.0363	0.1041	0.247
		INDEX	0.0502	0.0189	0.0148	0.0906
BSSMK → EKI → FIP	Conditional indirect effects	M-1SD	0.1065	0.029	0.0534	0.1667
		M + 1SD	0.2038	0.0459	0.1189	0.3014
		INDEX	0.0501	0.0249	0.0021	0.1005
BSSTK → EKI → FIP	Conditional indirect effects	M-1SD	0.1178	0.0303	0.058	0.1792
		M + 1SD	0.2205	0.0499	0.1261	0.3227
		INDEX	0.0529	0.0264	0.0027	0.1051

Source: Elaborated by authors

process. Moreover, this study further posits that environmental dynamics are essential contingency factors in explaining the changing relationship among boundary-spanning search, strategic knowledge integration and FIP. Through empirical validation of these relationships, this study reveals that the influence of boundary-spanning search on FIP is a complex process. Strategic knowledge integration is an essential factor contributing to FIP and mediates the influence of boundary-spanning on FIP. Furthermore, in high-level dynamic environments, the influence of strategic knowledge integration on FIP is amplified, as well as the impact of boundary-spanning search on FIP through strategic knowledge integration. Consequently, firms are urged to strengthen the roles of boundary-spanning search and strategic knowledge integration, particularly within high-level dynamic environments, to enhance innovation performance.

5.1 Theoretical implications

This study makes several theoretical contributions. First, it enhances our comprehension of the nature and implications of boundary-spanning search. Specifically, this study differentiates between boundary-spanning search and external search. Boundary-spanning search refers to organizations surpassing existing organizational, technological and geographical boundaries to acquire new knowledge and opportunities (Phene *et al.*, 2006). This involves determining which boundaries to cross when seeking external knowledge across organizations (Duan *et al.*, 2023), distinct from external search. This conceptual distinction refines the understanding that boundary-spanning search involves searching for external knowledge resources beyond the current organizational boundaries (Grimpe and Sofka, 2009; Leiponen and Helfat, 2010). Moreover, the study categorizes boundary-spanning search into two dimensions: boundary-spanning search for technical knowledge and search for market knowledge. Although previous studies have commonly classified boundary-spanning search based on search depth and breadth (Katila and Ahuja, 2002; Wang *et al.*, 2020a, 2020b; Rui and Lyytinen, 2019), they do not explicitly address search strategies related to innovative elements and boundary-spanning types. Additionally, this study highlights the significant role of boundary-spanning search in enabling firms to acquire heterogeneous knowledge, overcome their knowledge reserves, avoid maturity traps and broaden strategic portfolio spaces, thereby enhancing novelty (Hou *et al.*, 2023; Wang *et al.*, 2023; Yu *et al.*, 2019). These findings align with the knowledge-based view and extend previous research on the positive impact of external search on innovation performance (Schilling and Green, 2011; Rui and Lyytinen, 2019; Kneeland *et al.*, 2020; Gao *et al.*, 2022). Therefore, this study contributes to an enhanced theoretical understanding of boundary-spanning search.

Second, this study contributes to the literature on knowledge integration from the perspective of strategic knowledge management. It conceptualizes strategic knowledge integration as a two-dimensional construct and emphasizes its mediating role in boundary-spanning search and FIP. Previous research on knowledge integration has primarily focused on categorization based on knowledge sources, knowledge attributes, integration mechanisms and internal structural variances (De Boer *et al.*, 1999; Mitchell, 2006; Rialti *et al.*, 2020), as well as the comprehensive process of knowledge integration (Wang *et al.*, 2023). Although these studies have established a solid theoretical foundation for examining knowledge integration, they have not adequately explored the aspects of knowledge integration from the perspectives of firm planning and sustainable development. It is crucial to effectively integrate knowledge for firms' survival and development (Kamuriwo and Baden-Fuller, 2016; Gao *et al.*, 2021). The significance of strategic knowledge management is increasingly recognized, and the need for knowledge integration as a strategic tool necessitates the development of long-term plans to use and develop a firm's knowledge assets (Bolisani and Bratianu, 2017). However, previous research on strategic knowledge integration has primarily focused on a single dimension or remained at a theoretical level

(Hughes and Hodgkinson, 2021; David *et al.*, 2021; Marques *et al.*, 2016; Bolisani and Bratianu, 2017). To address this research gap, our study adopts Mintzberg's (1978) theory of strategic management classification and categorizes strategic knowledge integration into planned and emergent dimensions. This novel categorization aligns with Bolisani and Bratianu's (2017) proposition that knowledge is a crucial strategic decision resource for businesses and Guo *et al.*'s (2020) definition of strategic knowledge integration as the synthesis and utilization of information to achieve predefined strategic objectives or formulate new goals.

We further highlight strategic knowledge integration as an essential mediating variable in different dimensions through which boundary-spanning search influences innovation performance. While some researchers have demonstrated the positive influence of external knowledge search on a firm's innovation performance (Wang and Wang, 2022), the empirical analysis of the role of strategic knowledge integration in these studies has been limited, primarily focusing on knowledge-oriented approaches. Previous studies have emphasized that knowledge diversity can significantly enhance a firm's knowledge integration capability (Tzabbar *et al.*, 2013) and have validated the relationship between internal and external knowledge integration and new product innovation in firms (Homburg and Kuehnl, 2014). However, these studies have not adequately explored the "black box" of the internal relationship between boundary-spanning search and a firm's innovation performance (Zan *et al.*, 2024; Lyu *et al.*, 2023). Therefore, our study responds to the call for investigating the mediating effect of knowledge integration under the open innovation mode (Mell *et al.*, 2022). Our findings indicate that the abundant knowledge resources obtained through boundary-spanning search serve as a valuable resource pool for a firm's knowledge integration. Firms with high levels of knowledge integration can effectively assimilate new knowledge and develop knowledge routines that are difficult to imitate (Yang *et al.*, 2021), thus promoting the firm's innovation performance based on strategic knowledge integration. Our study not only expands on the proposal by Hou *et al.* (2023) to enhance a firm's innovation performance through unconventional search strategies but also partially supports the arguments made by Du *et al.* (2018), Salunke *et al.* (2019) and Yang *et al.* (2021) regarding the conversion of external knowledge into superior firm performance, which heavily depends on knowledge integration.

Third, this study establishes a theoretical framework encompassing boundary-spanning search, strategic knowledge integration and FIP, taking into account environmental dynamics. It clarifies the boundary conditions under which strategic knowledge integration exerts influence and comprehensively reveals the mechanism of boundary-spanning search on innovation performance. While previous research has primarily emphasized the unpredictability of environmental dynamics and its moderating effect on the relationship between internal resource capabilities and organizational performance (Wang *et al.*, 2011), there has been limited in-depth analysis of the boundary conditions of environmental dynamics in the process of strategic knowledge integration and FIP (David *et al.*, 2021), as well as a lack of research on the moderated mediation effects related to environmental dynamics. Environmental dynamics heightens the requirements for strategic knowledge integration, thus affecting the fit between strategic knowledge integration and the requirements. Given that environmental dynamics, as a crucial situational factor, significantly impacts firms' knowledge management activities, this study introduces environmental dynamics as a moderating variable and also considers its moderated mediation effects. The results indicate that environmental dynamics have a positive moderating effect on the relationship between the two dimensions of strategic knowledge integration and FIP, thus partially supporting the proposition in previous literature that under highly dynamic environments, external knowledge search can exert a more positive influence on organizational creativity (Men *et al.*, 2020; Liu, 2021). This study also illustrates that a highly dynamic environment positively moderates the mediating effect of boundary-spanning search through strategic knowledge integration on FIP. Therefore, this study

comprehensively elucidates the mechanism of boundary-spanning search on innovation performance, offering new insights into the literature on FIP and enriching the overall theoretical framework.

5.2 Managerial implications

This study carries significant implications for practical applications. First, the findings of our study confirm the significance of boundary-spanning search as a crucial driver of FIP. In the current knowledge-intensive and fiercely competitive business environment, firms must adopt boundary-spanning search strategies to enhance innovation performance. On the one hand, firms should actively strengthen collaboration with universities, research institutions and other external entities (Ryu *et al.*, 2022). Through such collaboration, firms can acquire cutting-edge technological knowledge and integrate it with internal experience, continuously excavating and integrating advanced knowledge to promote internal technological transformation and develop unique innovative knowledge. This specialized BSSTK facilitates the integration and application of technological knowledge within firms, driving the advancement of innovation (Hoppmann *et al.*, 2021). On the other hand, firms should fully use BSSMK. By collaborating with customers, upstream and downstream supply chain companies, and competitors, firms can obtain information on market demands, positioning and sales channels (Cui *et al.*, 2022). Through the establishment of collaborative mechanisms, firms can forge new sales channels and provide personalized customized services, thereby offering rich market knowledge resources for the firm's innovation process.

Next, this study also reveals the vital role of strategic knowledge integration in promoting the influence of boundary-spanning search on FIP. The knowledge derived from boundary-spanning search is often fragmented, and it is only through meaningful combinations that this knowledge can create strategic value (Alvarez and Barney, 2007; Zahra, 2015). Managers should fully recognize the instrumental role of knowledge integration from a strategic management perspective (David *et al.*, 2021). On the one hand, managers should use planned strategic knowledge integration to bridge the knowledge gaps that hinder the implementation of existing strategies, thus achieving predetermined strategic objectives. On the other hand, managers should adopt emergent strategic knowledge integration to update their understanding of the internal and external environment (Zeigermann, 2021), and redefine their strategic goals. By coordinating strategic knowledge management, organizations can transform from passive to active, enabling them to rapidly introduce innovative products and seize market opportunities.

Finally, our findings emphasize the critical role of environmental dynamics as a key contingency factor influencing the relationship among boundary-spanning search, strategic knowledge integration and FIP. This implies that firms should consider implementing efficient dynamic knowledge management. The speed and scope of market fluctuations often entail uncertainty, necessitating organizational agility and the implementation of dynamic knowledge management (Lakshman *et al.*, 2023). In high-dynamic environments, firms should rapidly adjust their strategic deployment, proactively monitor and capture market dynamics (Kim and Lim, 2022), update existing knowledge systems and paradigms and drive rapid output of innovative outcomes through effective boundary-spanning search and targeted strategic knowledge integration. On the contrary, in low-dynamic environments, firms can appropriately reduce the costs associated with boundary-spanning search and strategic knowledge integration management to attain greater innovation performance.

5.3 Limitations and future research

This study has some limitations to be addressed for future research. First, the adoption of a cross-sectional research design in this work presents challenges in establishing causation

in the examined relationships. To overcome this limitation, future studies are encouraged to use a longitudinal method and collect data across multiple stages, allowing for a better understanding of causal relationships over time. Second, the sample used is from China. Although the conclusions offer valuable insights for guiding Chinese firms, they may not be universally applicable. Therefore, future research efforts should encompass diverse countries and cultural backgrounds to enhance the generalizability of the study. Third, this study investigates the influence of boundary-spanning search for market and technological knowledge on FIP. However, the concept of innovation performance may be too broad to capture the distinct classification results of these two types of searches. For instance, innovation performance can be categorized into product innovation and process innovation. To compare the influences of boundary-spanning search on different types of innovation performance, future research could consider incorporating a second-order structure that allows for a more nuanced examination of FIP.

Except for these limitations, two additional directions should be noted in future research. First, this study investigates the influence of planned and emergent strategic knowledge integration on the relationship between boundary-spanning search and FIP. However, other types of strategic knowledge integration could be explored. For example, research has shown that simultaneous exploration and exploitation activities, as proposed by ambidexterity theory, are crucial for firm success (March, 1991; Luo *et al.*, 2019; Li *et al.*, 2022). Therefore, future research could incorporate the ambidexterity theory to investigate the effects of both exploitative and exploratory strategic knowledge integration on the relationship between boundary-spanning search and FIP. Second, the importance of digital technological capabilities has been increasingly recognized, as they play a crucial role in perceiving emerging opportunities and expanding the scope of firm knowledge management (Mikalef and Krogstie, 2020; Gao *et al.*, 2022; Shi *et al.*, 2023). Therefore, it is important to explore the effect of technological capabilities on boundary-spanning search and FIP.

References

- Albawwat, I.E. (2022), "Tacit knowledge sharing in small audit firms and audit quality inputs: the antecedent effect of auditors' social capital", *Journal of Knowledge Management*, Vol. 26 No. 9, pp. 2333-2353.
- Alvarez, S.A. and Barney, J.B. (2005), "How do entrepreneurs organize firms under conditions of uncertainty?", *Journal of Management*, Vol. 31 No. 5, pp. 776-793.
- Alvarez, S.A. and Barney, J.B. (2007), "Discovery and creation: alternative theories of entrepreneurial action", *Strategic Entrepreneurship Journal*, Vol. 1 Nos 1/2, pp. 11-26.
- An, W., Huang, Q., Liu, H. and Wu, J. (2022), "The match between business model design and knowledge base in firm growth: from a knowledge-based view", *Technology Analysis & Strategic Management*, Vol. 34 No. 1, pp. 99-111.
- Archer-Brown, C. and Kietzmann, J. (2018), "Strategic knowledge management and enterprise social media", *Journal of Knowledge Management*, Vol. 22 No. 6, pp. 1288-1309.
- Armstrong, J.S. and Overton, T.S. (1977), "Estimating nonresponse bias in mail surveys", *Journal of Marketing Research*, Vol. 14 No. 3, pp. 396-402.
- Asimakopoulous, G., Revilla, A.J. and Slavova, K. (2020), "External knowledge sourcing and firm innovation efficiency", *British Journal of Management*, Vol. 31 No. 1, pp. 123-140.
- Bagozzi, R.P., Yi, Y. and Phillips, L.W. (1991), "Assessing construct validity in organizational research", *Administrative Science Quarterly*, Vol. 36 No. 3, pp. 421-458.
- Baron, R.M. and Kenny, D.A. (1986), "The moderator-mediator variable distinction in social psychological research: conceptual, strategic, and statistical considerations", *Journal of Personality and Social Psychology*, Vol. 51 No. 6, pp. 1173-1182.
- Blythe, J.N. and Dadi, U. (2012), "Knowledge integration as a method to develop capacity for evaluating technical information on biodiversity and ocean currents for integrated coastal management", *Environmental Science & Policy*, Vol. 19-20, pp. 49-58.

- Bolisani, E. and Bratianu, C. (2017), "Knowledge strategy planning: an integrated approach to manage uncertainty, turbulence, and dynamics", *Journal of Knowledge Management*, Vol. 21 No. 2, pp. 233-253.
- Brettel, M., Mauer, R., Engelen, A. and Küpper, D. (2012), "Corporate effectuation: entrepreneurial action and its impact on R&D project performance", *Journal of Business Venturing*, Vol. 27 No. 2, pp. 167-184.
- Chandler, G.N., DeTienne, D.R., McKelvie, A. and Mumford, T.V. (2011), "Causation and effectuation processes: a validation study", *Journal of Business Venturing*, Vol. 26 No. 3, pp. 375-390.
- Chang, S.J., Van Witteloostuijn, A. and Eden, L. (2010), "From the editors: common method variance in international business research", *Journal of International Business Studies*, Vol. 41 No. 2, pp. 178-184.
- Chen, J., Zhao, Y., Wang, Y. and Jia, T. (2023), "How boundary-spanning search brings about innovation performance: the contingent effects of in house training of employees and communication investment", *Technology Analysis & Strategic Management*, doi: [10.1080/09537325.2023.2222197](https://doi.org/10.1080/09537325.2023.2222197).
- Cheng, C.C. and Shiu, E.C. (2015), "The inconvenient truth of the relationship between open innovation activities and innovation performance", *Management Decision*, Vol. 53 No. 3, pp. 625-647.
- Chesbrough, H., Lettl, C. and Ritter, T. (2018), "Value creation and value capture in open innovation", *Journal of Product Innovation Management*, Vol. 35 No. 6, pp. 930-938.
- Cohen, W.M. and Levinthal, D.A. (1990), "Absorptive capacity: a new perspective on learning and innovation", *Administrative Science Quarterly*, Vol. 35 No. 1, pp. 128-152.
- Collis, J. and Hussey, R. (2013), *Business Research: A Practical Guide for Undergraduate and Postgraduate Students*, Macmillan International Higher Education.
- Cui, T., Tong, Y. and Tan, C.H. (2022), "Open innovation and information technology use: towards an operational alignment view", *Information Systems Journal*, Vol. 32 No. 5, pp. 932-972.
- Cyert, R.M. and March, J.G. (1963), *A Behavioral Theory of the Firm*, Prentice-Hall.
- David, K.G., Wu, Y., Pei, C. and Moosa, A. (2021), "The effects of top management team external social capital on firm's openness: exploring the mediating role of strategic knowledge integration", *IEEE Transactions on Engineering Management*, Vol. 70 No. 11, pp. 3918-3931.
- De Boer, M., Van Den Bosch, F.A.J. and Volberda, H.W. (1999), "Managing organizational knowledge integration in the emerging multimedia complex", *Journal of Management Studies*, Vol. 36 No. 3, pp. 379-398.
- Donaldson, L. (2001), *The Contingency Theory Organizations*, Sage Publications.
- Du, L., Zhang, Z. and Feng, T. (2018), "Linking green customer and supplier integration with green innovation performance: the role of internal integration", *Business Strategy and the Environment*, Vol. 27 No. 8, pp. 1583-1595.
- Duan, K., Zhang, C., Li, J., Zhang, R. and Zhang, Y. (2020), "Boundary-spanning search for knowledge, knowledge reconstruction and the sustainable innovation ability of agricultural enterprises: a Chinese perspective", *Agriculture*, Vol. 10 No. 2, p. 39.
- Duan, Y., Liu, S., Mu, C., Liu, X., Cheng, E. and Liu, Y. (2023), "The moderating effect of managerial discretion on cross-border knowledge search and the innovation quality of high-tech firms in a global health emergency: evidence from China", *Journal of Knowledge Management*, Vol. 27 No. 1, pp. 121-155.
- Dubey, R., Bryde, D.J., Blome, C., Roubaud, D. and Giannakis, M. (2021), "Facilitating artificial intelligence powered supply chain analytics through alliance management during the pandemic crises in the B2B context", *Industrial Marketing Management*, Vol. 96, pp. 135-146.
- Forés, B. and Camisón, C. (2016), "Does incremental and radical innovation performance depend on different types of knowledge accumulation capabilities and organizational size?", *Journal of Business Research*, Vol. 69 No. 2, pp. 831-848.
- Gao, H., Ding, X.H. and Wu, S. (2022), "Impact of knowledge search on product and process innovation: mediating role of absorptive capacity and moderating role of IT capability", *European Journal of Innovation Management*, Vol. 25 No. 2, pp. 325-346.
- Gao, J., He, H., Teng, D., Wan, X. and Zhao, S. (2021), "Cross-border knowledge search and integration mechanism-a case study of Haier Open Partnership Ecosystem (HOPE)", *Chinese Management Studies*, Vol. 15 No. 2, pp. 428-455.
- Geroski, P., Machin, S. and Van Reenen, J. (1993), "The profitability of innovating firms", *The RAND Journal of Economics*, Vol. 24 No. 2, pp. 198-211.

- Grant, R.M. (1996), "Toward a knowledge-based theory of the firm", *Journal of Strategy and Management*, Vol. 17 No. S2, pp. 109-122.
- Grimpe, C. and Sofka, W. (2009), "Search patterns and absorptive capacity: low and high technology sectors in European countries", *Research Policy*, Vol. 38 No. 3, pp. 495-506.
- Guan, J.C., Richard, C.M., Tang, E.P. and Lau, A.K. (2009), "Innovation strategy and performance during economic transition: evidences in Beijing, China", *Research Policy*, Vol. 38 No. 5, pp. 802-812.
- Guo, R., Cai, L. and Fei, Y. (2019), "Knowledge integration methods, product innovation and high-tech new venture performance in China", *Technology Analysis & Strategic Management*, Vol. 31 No. 3, pp. 306-318.
- Guo, R., Lv, X., Wang, Y., Chaudhry, P.E. and Chaudhry, S.S. (2020), "Decision-making logics and high-tech entrepreneurial opportunity identification: the mediating role of strategic knowledge integration", *Systems Research and Behavioral Science*, Vol. 37 No. 4, pp. 719-733.
- Hayes, A.F. (2009), "Beyond Baron and Kenny: statistical mediation analysis in the new millennium", *Communication Monographs*, Vol. 76 No. 4, pp. 408-420.
- Homburg, C. and Kuehnl, C. (2014), "Is the more always better? A comparative study of internal and external integration practices in new product and new service development", *Journal of Business Research*, Vol. 67 No. 7, pp. 1360-1367.
- Hoppmann, J., Wu, G. and Johnson, J. (2021), "The impact of demand-pull and technology-push policies on firms' knowledge search", *Technological Forecasting and Social Change*, Vol. 170, p. 120863.
- Hou, T., Li, J.J. and Lin, J. (2023), "Linking knowledge search to knowledge creation: the intermediate role of knowledge complexity", *Management Decision*, Vol. 61 No. 5, pp. 1156-1182.
- Huang, J.W. and Li, Y.H. (2017), "Green innovation and performance: the view of organizational capability and social reciprocity", *Journal of Business Ethics*, Vol. 145 No. 2, pp. 309-324.
- Hughes, P. and Hodgkinson, I. (2021), "Knowledge management activities and strategic planning capability development", *European Business Review*, Vol. 33 No. 2, pp. 238-254.
- Jansen, J., J.P., Van Den Bosch, F.A.J. and Volberda, H.W. (2005), "Managing potential and realized absorptive capacity: how do organizational antecedents matter?", *Academy of Management Journal*, Vol. 48 No. 6, pp. 999-1015.
- Jin, S., Wang, J. and Zhu, P. (2024), "Environmental scanning, resource orchestration, and disruptive innovation", *R&D Management*, doi: [10.1111/radm.12681](https://doi.org/10.1111/radm.12681).
- Kamuriwo, D.S. and Baden-Fuller, C. (2016), "Knowledge integration using product R&D outsourcing in biotechnology", *Research Policy*, Vol. 45 No. 5, pp. 1031-1045.
- Katila, R. and Ahuja, G. (2002), "Something old, something new: a longitudinal study of search behavior and new product introduction", *Academy of Management Journal*, Vol. 45 No. 6, pp. 1183-1194.
- Kim, K. and Lim, G. (2022), "International dynamic marketing capabilities of emerging-market small business on e-commerce", *Journal of Theoretical and Applied Electronic Commerce Research*, Vol. 17 No. 1, pp. 199-211.
- Kneeland, M.K., Schilling, M.A. and Aharonson, B.S. (2020), "Exploring uncharted territory: knowledge search processes in the origination of outlier innovation", *Organization Science*, Vol. 31 No. 3, pp. 535-557.
- Kumar, S., Raut, R.D., Priyadarshinee, P., Mangla, S.K., Awan, U. and Narkhede, B.E. (2022), "The impact of IoT on the performance of vaccine supply chain distribution in the COVID-19 context", *IEEE Transactions on Engineering Management*, doi: [10.1109/TEM.2022.3157625](https://doi.org/10.1109/TEM.2022.3157625).
- Laihonen, H. and Mäntylä, S. (2018), "Strategic knowledge management and evolving local government", *Journal of Knowledge Management*, Vol. 22 No. 1, pp. 219-234.
- Lakshman, C., Lakshman, S. and Gok, K. (2023), "Managers' knowledge and customer-focused knowledge management as dynamic capabilities: implications for innovation performance", *Asian Business & Management*, Vol. 22 No. 1, pp. 246-274.
- Laursen, K. and Salter, A. (2006), "Open for innovation: the role of openness in explaining innovation performance among U.K. manufacturing firms", *Strategic Management Journal*, Vol. 27 No. 2, pp. 131-150.
- Leiponen, A. and Helfat, C.E. (2010), "Innovation objectives, knowledge sources, and the benefits of breadth", *Strategic Management Journal*, Vol. 31 No. 2, pp. 224-236.

- Lemon, M. and Sahota, P.S. (2004), "Organizational culture as a knowledge repository for increased innovative capacity", *Technovation*, Vol. 24 No. 6, pp. 483-498.
- Leonard-Barton, D. (1992), "Core capabilities and core rigidities: a paradox in managing new product development", *Strategic Management Journal*, Vol. 13 No. S1, pp. 111-125.
- Levitt, B. and March, J.G. (1988), "Organizational learning", *Annual Review of Sociology*, Vol. 14 No. 1, pp. 319-340.
- Li, X. and Li, K. (2023), "Knowledge networks and ambidextrous learning: what is the impact on innovation performance?", *Revista de Administração de Empresas*, Vol. 63 No. 6, p. e2022-0105.
- Li, X., Li, K. and Zhou, H. (2022), "Impact of inventor's cooperation network on ambidextrous innovation in Chinese AI enterprises", *Sustainability*, Vol. 14 No. 16, p. 9996.
- Li, Y., Vanhaverbeke, W. and Schoenmakers, W. (2008), "Exploration and exploitation in innovation: reframing the interpretation", *Creativity and Innovation Management*, Vol. 17 No. 2, pp. 107-126.
- Liu, B. (2021), "Matching external search strategies with radical and incremental innovation and the role of knowledge integration capability", *Baltic Journal of Management*, Vol. 16 No. 5, pp. 765-784.
- Liu, F.C., Luo, L. and Zhang, S.H. (2021), "Knowledge attributes, knowledge relations and firms' innovation performance in the R&D cooperation", *Science Research Management*, Vol. 42 No. 11, pp. 155-163.
- Lu, H.T., Li, X. and Yuen, K.F. (2023), "Digital transformation as an enabler of sustainability innovation and performance-Information processing and innovation ambidexterity perspectives", *Technological Forecasting and Social Change*, Vol. 196, p. 122860.
- Luo, C., Kumar, S., Mallick, D.N. and Luo, B. (2019), "Impacts of exploration and exploitation on firm's performance and the moderating effects of slack: a panel data analysis", *IEEE Transactions on Engineering Management*, Vol. 66 No. 4, pp. 613-620.
- Lyu, T., Geng, Q. and Zhao, Q. (2023), "Understanding the efforts of cross-border search and knowledge co-creation on manufacturing enterprises' service innovation performance", *Systems*, Vol. 11 No. 1, p. 4.
- Lyu, C., Peng, C., Yang, H., Li, H. and Gu, X. (2022), "Social capital and innovation performance of digital firms: serial mediation effect of cross-border knowledge search and absorptive capacity", *Journal of Innovation & Knowledge*, Vol. 7 No. 2, p. 100187.
- McIver, D. and Lepisto, D.A. (2017), "Effects of knowledge management on unit performance: examining the moderating role of tacitness and learnability", *Journal of Knowledge Management*, Vol. 21 No. 4, pp. 796-816.
- March, J.G. (1991), "Exploration and exploitation in organizational learning", *Organization Science*, Vol. 2 No. 1, pp. 71-87.
- Marques, C.S., Leal, C., Marques, C.P. and Cardoso, A.R. (2016), "Strategic knowledge management, innovation and performance: a qualitative study of the footwear industry", *Journal of the Knowledge Economy*, Vol. 7 No. 3, pp. 659-675.
- Mell, J.N., Van Knippenberg, D., Van Ginkel, W.P. and Heugens, P.P. (2022), "From boundary spanning to intergroup knowledge integration: the role of boundary spanners' metaknowledge and proactivity", *Journal of Management Studies*, Vol. 59 No. 7, pp. 1723-1755.
- Men, C., Fong, P.S.W., Luo, J., Zhong, J. and Huo, W. (2019), "When and how knowledge sharing benefits team creativity: the importance of cognitive team diversity", *Journal of Management & Organization*, Vol. 25 No. 6, pp. 807-824.
- Mikalef, P. and Krogstie, J. (2020), "Examining the interplay between big data analytics and contextual factors in driving process innovation capabilities", *European Journal of Information Systems*, Vol. 29 No. 3, pp. 260-287.
- Mintzberg, H. (1978), "Patterns in strategy formation", *Management Science*, Vol. 24 No. 9, pp. 934-948.
- Mitchell, V.L. (2006), "Knowledge integration and information technology project performance", *Mis Quarterly*, Vol. 30 No. 4, pp. 919-939.
- Nelson, R.R. and Winter, S.G. (1982), *An Evolutionary Theory of Economic Change*, Belknap Press of Harvard University Press.
- Nerkar, A. and Roberts, P.W. (2004), "Technological and product-market experience and the success of new product introductions in the pharmaceutical industry", *Strategic Management Journal*, Vol. 25 Nos 8/9, pp. 779-799.

- Pati, R.K., Nandakumar, M.K., Ghobadian, A., Ireland, R.D. and O'Regan, N. (2018), "Business model design-performance relationship under external and internal contingencies: evidence from SMEs in an emerging economy", *Long Range Planning*, Vol. 51 No. 5, pp. 750-769.
- Phene, A., Fladmoe-Lindquist, K. and Marsh, L. (2006), "Breakthrough innovations in the US biotechnology industry: the effects of technological space and geographic origin", *Strategic Management Journal*, Vol. 27 No. 4, pp. 369-388.
- Podsakoff, P.M. and Organ, D.W. (1986), "Self-reports in organizational research: problems and prospects", *Journal of Management*, Vol. 12 No. 4, pp. 531-544.
- Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y. and Podsakoff, N.P. (2003), "Common method biases in behavioral research: a critical review of the literature and recommended remedies", *Journal of Applied Psychology*, Vol. 88 No. 5, pp. 879-903.
- Preacher, K.J. and Hayes, A.F. (2008), "Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models", *Behavior Research Methods*, Vol. 40 No. 3, pp. 879-891.
- Priem, R.L., Li, S. and Carr, J.C. (2012), "Insights and new directions from demand-side approaches to technology innovation, entrepreneurship, and strategic management research", *Journal of Management*, Vol. 38 No. 1, pp. 346-374.
- Qu, Y. and Mardani, A. (2023), "Market orientation, technological opportunity, and new product innovation performance", *Journal of Business Research*, Vol. 162, p. 113841.
- Reymen, I.M., Andries, P., Berends, H., Mauer, R., Stephan, U. and Van Burg, E. (2015), "Understanding dynamics of strategic decision making in venture creation: a process study of effectuation and causation", *Strategic Entrepreneurship Journal*, Vol. 9 No. 4, pp. 351-379.
- Rialti, R., Marzi, G., Caputo, A. and Mayah, K.A. (2020), "Achieving strategic flexibility in the era of big data: the importance of knowledge management and ambidexterity", *Management Decision*, Vol. 58 No. 8, pp. 1585-1600.
- Rindfleisch, A., Malter, A.J., Ganesan, S. and Moorman, C. (2008), "Cross-sectional versus longitudinal survey research: concepts, findings, and guidelines", *Journal of Marketing Research*, Vol. 45 No. 3, pp. 261-279.
- Romero, I. and Martínez-Román, J.A. (2012), "Self-employment and innovation. Exploring the determinants of innovative behavior in small businesses", *Research Policy*, Vol. 41 No. 1, pp. 178-189.
- Rosenkopf, L. and Nerkar, A. (2001), "Beyond local search: boundary-spanning, exploration, and impact in the optical disk industry", *Strategic Management Journal*, Vol. 22 No. 4, pp. 287-306.
- Rui, Z. and Lyytinen, K. (2019), "How do ventures become more innovative? The effect of external search and ambidextrous knowledge integration", *European Journal of Innovation Management*, Vol. 22 No. 5, pp. 845-865.
- Ryu, D., Baek, K.H. and Yoon, J. (2022), "Linking external knowledge search to innovation ambidexterity in SMEs", *Science, Technology and Society*, Vol. 27 No. 2, pp. 159-171.
- Sadeghi, V.J., Nkongolo-Bakenda, J.M., Anderson, R.B. and Dana, L.P. (2019), "An institution-based view of international entrepreneurship: a comparison of context-based and universal determinants in developing and economically advanced countries", *International Business Review*, Vol. 28 No. 6, p. 101588.
- Salunke, S., Weerawardena, J. and McColl-Kennedy, J.R. (2019), "The Central role of knowledge integration capability in service innovation-based competitive strategy", *Industrial Marketing Management*, Vol. 76, pp. 144-156.
- Santoro, G., Bresciani, S. and Papa, A. (2020), "Collaborative modes with cultural and creative industries and innovation performance: the moderating role of heterogeneous sources of knowledge and absorptive capacity", *Technovation*, Vol. 92-93, p. 102040.
- Sarasvathy, S.D. (2001), "Causation and effectuation: toward a theoretical shift from economic inevitability to entrepreneurial contingency", *The Academy of Management Review*, Vol. 26 No. 2, pp. 243-263.
- Schilke, O. (2014), "On the contingent value of dynamic capabilities for competitive advantage: the nonlinear moderating effect of environmental dynamism", *Strategic Management Journal*, Vol. 35 No. 2, pp. 179-203.
- Schilling, M.A. and Green, E. (2011), "Recombinant search and breakthrough idea generation: an analysis of high impact papers in the social sciences", *Research Policy*, Vol. 40 No. 10, pp. 1321-1331.

- Segarra-Ciprés, M. and Bou-Lluisar, J.C. (2018), "External knowledge search for innovation: the role of firms' innovation strategy and industry context", *Journal of Knowledge Management*, Vol. 22 No. 2, pp. 280-298.
- Shao, Y., Bi, S., Liu, P. and Chen, J. (2023), "The effect of boundary-spanning search and cognitive style on the business model design of internet start-ups: evidence from China", *Technology Analysis & Strategic Management*, doi: [10.1080/09537325.2023.2250015](https://doi.org/10.1080/09537325.2023.2250015).
- Shi, H., Feng, T. and Zhu, Z. (2023), "The impact of big data analytics capability on green supply chain integration: an organizational information processing theory perspective", *Business Process Management Journal*, Vol. 29 No. 2, pp. 550-577.
- Sidhu, J.S., Volberda, H.W. and Commandeur, H.R. (2004), "Exploring exploration orientation and its determinants: some empirical evidence", *Journal of Management Studies*, Vol. 41 No. 6, pp. 913-932.
- Sidhu, J.S., Commandeur, H.R. and Volberda, H.W. (2007), "The multifaceted nature of exploration and exploitation: value of supply, demand and spatial search for innovation", *Organization Science*, Vol. 18 No. 1, pp. 20-38.
- Singh, J. (1995), "Measurement issues in cross-national research", *Journal of International Business Studies*, Vol. 26 No. 3, pp. 597-619.
- Slevin, D.P. and Covin, J.G. (1997), "Strategy formation patterns, performance, and the significance of context", *Journal of Management*, Vol. 23 No. 2, pp. 189-209.
- Stuart, T. and Podolny, J. (1996), "Local search and the evolution of technological capabilities", *Strategic Management Journal*, Vol. 17 No. S1, pp. 21-38.
- Terjesen, S. and Patel, P.C. (2017), "In search of process innovations: the role of search depth, search breadth, and the industry environment", *Journal of Management*, Vol. 43 No. 5, pp. 1421-1446.
- Titus, V.K., Jr, Covin, J.G. and Slevin, D.P. (2011), "Aligning strategic processes in pursuit of firm growth", *Journal of Business Research*, Vol. 64 No. 5, pp. 446-453.
- Todorova, G. and Durisin, B. (2007), "Absorptive capacity: valuing a reconceptualization", *Academy of Management Review*, Vol. 32 No. 3, pp. 774-786.
- Tzabbar, D., Aharonson, B.S. and Amburgey, T.L. (2013), "When does tapping external sources of knowledge result in knowledge integration?", *Research Policy*, Vol. 42 No. 2, pp. 481-494.
- Ul-Durar, S., Awan, U., Varma, A., Memon, S. and Mention, A.L. (2023), "Integrating knowledge management and orientation dynamics for organization transition from eco-innovation to circular economy", *Journal of Knowledge Management*, Vol. 27 No. 8, pp. 2217-2248.
- Van de Ven, A.H. (1986), "Central problems in the management of innovation", *Management Science*, Vol. 32 No. 5, pp. 590-607.
- Vera, D. and Crossan, M. (2005), "Improvisation and innovative performance in teams", *Organization Science*, Vol. 16 No. 3, pp. 203-224.
- Wang, M. and Wang, H. (2022), "Knowledge search and innovation performance: the mediating role of absorptive capacity", *Operations Management Research*, Vol. 15 Nos 3/4, pp. 1060-1070.
- Wang, C., Chin, T. and Lin, J.H. (2020a), "Openness and firm innovation performance: the moderating effect of ambidextrous knowledge search strategy", *Journal of Knowledge Management*, Vol. 24 No. 2, pp. 301-323.
- Wang, J., Xue, Y. and Yang, J. (2020b), "Boundary-spanning search and firms' green innovation: moderating role of resource orchestration capability", *Business Strategy and the Environment*, Vol. 29 No. 2, pp. 361-374.
- Wang, L., Wang, W. and Li, Q. (2023), "Research on the impact of boundary-spanning search on the sustainable development ability of service-oriented manufacturing enterprises: evidence from Chinese listed companies", *Journal of Organizational Change Management*, Vol. 36 No. 1, pp. 3-20.
- Wang, L., Yeung, J.H.Y. and Zhang, M. (2011), "The impact of trust and contract on innovation performance: the moderating role of environmental uncertainty", *International Journal of Production Economics*, Vol. 134 No. 1, pp. 114-122.
- Wang, D., Song, J., Sun, X. and Wang, X. (2022), "A study on the impact of boundary-spanning search on the sustainable development performance of technology start-ups", *Sustainability*, Vol. 14 No. 15, p. 9182.
- Wang, G., Dou, W., Zhu, W. and Zhou, N. (2015), "The effects of firm capabilities on external collaboration and performance: the moderating role of market turbulence", *Journal of Business Research*, Vol. 68 No. 9, pp. 1928-1936.

- Wang, Y., Jia, T., Chen, J. and Sun, H. (2019), "Recombine supplier-side search via innovation ambidexterity: an empirical study on Hong Kong manufacturing firms", *International Journal of Physical Distribution & Logistics Management*, Vol. 49 No. 2, pp. 178-199.
- Wang, N., Gong, Z., Xu, Z., Liu, Z. and Han, Y. (2021), "A quantitative investigation of the technological innovation in large construction companies", *Technology in Society*, Vol. 65, p. 101533.
- Wei, Y.S. and Atuahene-Gima, K. (2009), "The moderating role of reward systems in the relationship between market orientation and new product performance in China", *International Journal of Research in Marketing*, Vol. 26 No. 2, pp. 89-96.
- West, J., Salter, A., Vanhaverbeke, W. and Chesbrough, H. (2014), "Open innovation: the next decade", *Research Policy*, Vol. 43 No. 5, pp. 805-811.
- Williams, L.J. and Anderson, S.E. (1994), "An alternative approach to method effects by using Latent-Variable models: applications in organizational behavior research", *Journal of Applied Psychology*, Vol. 79 No. 3, pp. 323-331.
- Williams, L.J., Gavin, M.B. and Williams, M.L. (1996), "Measurement and nonmeasurement processes with negative affectivity and employee attitudes", *Journal of Applied Psychology*, Vol. 81 No. 1, pp. 88-101.
- Wu, W., Huang, X., Wu, C.H. and Tsai, S.B. (2021), "Technological boundary-spanning search, crowdfunding interaction and crowdfunding innovation performance: a mediated moderation model of knowledge sharing", *Enterprise Information Systems*, Vol. 15 No. 3, pp. 352-372.
- Xu, J., Li, F., Li, L. and Hou, J. (2017), "The effect of external relationship quality on low carbon technology innovation: knowledge perspective", *China Soft Science*, No. 2, pp. 183-192.
- Yang, M., Wang, J. and Yang, J. (2021), "Boundary-spanning search, knowledge integration capability and sustainable competitive advantage", *Baltic Journal of Management*, Vol. 16 No. 3, pp. 446-464.
- Ye, F., Yang, Y., Xia, H., Shao, Y., Gu, X. and Shen, J. (2022), "Green entrepreneurial orientation, boundary-spanning search and enterprise sustainable performance: the moderating role of environmental dynamism", *Frontiers in Psychology*, Vol. 13, p. 978274.
- Yu, Y., Yuan, L. and Li, J. (2019), "Knowledge search modes and innovation performance: the moderating role of strategic R&D orientation", *Technology Analysis & Strategic Management*, Vol. 31 No. 6, pp. 625-640.
- Yuan, C., Xue, D. and He, X. (2021), "A balancing strategy for ambidextrous learning, dynamic capabilities, and business model design, the opposite moderating effects of environmental dynamism", *Technovation*, Vol. 103, p. 102225.
- Zack, M. (1999), "Developing a knowledge strategy", *California Management Review*, Vol. 41 No. 3, pp. 125-145.
- Zahra, S.A. (2015), "Corporate entrepreneurship as knowledge creation and conversion: the role of entrepreneurial hubs", *Small Business Economics*, Vol. 44 No. 4, pp. 727-735.
- Zan, A., Yao, Y. and Chen, H. (2024), "Knowledge search and firm innovation: the roles of knowledge inertia and knowledge integration capability", *Technology Analysis & Strategic Management*, Vol. 36 No. 6, pp. 1150-1165.
- Ze, R., Kun, Z., Boadu, F. and Yu, L. (2018), "The effects of boundary-spanning search, network ties, and absorptive capacity for innovation: a moderated mediation examination", *Sustainability*, Vol. 10 No. 11, p. 3980.
- Zeigermann, U. (2021), "Scientific knowledge integration and the implementation of the SDGs: comparing strategies of sustainability networks", *Politics and Governance*, Vol. 9 No. 1, pp. 164-175.
- Zhang, F., Zhu, L. and Lyu, C. (2022), "Geographic search, potential absorptive capacity and radical innovation performance: the moderating role of collaborative network", *European Journal of Innovation Management*, Vol. 25 No. 2, pp. 479-500.
- Zhang, H. and Hu, B. (2017), "The effects of organizational isomorphism on innovation performance through knowledge search in industrial cluster", *Chinese Management Studies*, Vol. 11 No. 2, pp. 209-229.
- Zhang, W., Zhao, Y., Tian, L. and Liu, D. (2017), "Boundary-spanning demand-side search and radical technological innovations in China: the moderation of innovation appropriability", *Management Decision*, Vol. 55 No. 8, pp. 1749-1769.

Zhang, W., Zhao, Y., Wang, D., Wang, H. and Li, J. (2019), "Ambidextrous search and product innovation: moderating effects of resource and structural attributes", *The Journal of Technology Transfer*, Vol. 44 No. 4, pp. 1007-1028.

Zhao, Y., Feng, T. and Shi, H. (2018), "External involvement and green product innovation: the moderating role of environmental uncertainty", *Business Strategy and the Environment*, Vol. 27 No. 8, pp. 1167-1180.

Zhou, K.Z. and Li, C.B. (2012), "How knowledge affects radical innovation: knowledge base, market knowledge acquisition, and internal knowledge sharing", *Strategic Management Journal*, Vol. 33 No. 9, pp. 1090-1102.

Zhu, P., Miao, X. and Jin, S. (2024), "Boundary-spanning search and business model innovation: the joint moderating effects of innovative cognitive imprinting and environmental dynamics", *European Journal of Innovation Management*, Vol. 27 No. 1, pp. 310-333.

Zhu, P., Miao, X., Jin, S. and Moehler, R. (2023), "Transactive memory system, boundary-spanning search and business model innovation: the moderating role of environmental dynamism", *Technovation*, Vol. 128, p. 102852.

Further reading

Men, C., Luo, J., Fong, P.S., Zhong, J. and Huo, W. (2020), "Translating external knowledge to team creativity in turbulent environments: the role of absorptive capacity and knowledge integration", *The Journal of Creative Behavior*, Vol. 54 No. 2, pp. 363-377.

Table A1 Survey item

Measurement items	Source
<i>Construct 1: boundary-spanning search for market knowledge (BSSMK)</i>	Sidhu et al. (2007)
1a. We closely monitors the dynamics of companies with whom we have established cooperative relationships	Duan et al. (2020)
1b. We proactively seeks to understand changes in customer preferences for products or services	
1c. We closely monitors the dynamics of upstream and downstream companies in the supply chain	
1d. We collect information on potential customer preferences for products or services	
1e. We pay attention to activities of companies that provide complementary products	
1f. We have an in-depth understanding of our customers' product and process innovation endeavors.	
1g. We pay attention to industries that have different business scopes but still cater to similar customers	
<i>Construct 2: boundary-spanning search for technological knowledge (BSSTK)</i>	Sidhu et al. (2007)
2a. We keep track of technological developments in the current industry	Zhang et al. (2019)
2b. We monitor the dynamics of companies that are similar to us in terms of technology	Duan et al. (2020)
2c. We regularly update our technologies	
2d. We consider how to use existing production equipment for manufacturing new products	
2e. We pay attention to whether competing companies have updated their production technologies	
2f. We focus on technological advancements in the processes of products or services provided by our suppliers	
2g. We gather information on industries with similar technology as well as firms that possess comparable technological capabilities	
<i>Construct 3: planned strategic knowledge integration (PKI)</i>	Marques et al. (2016)
3a. We formulate learning plans based on existing knowledge to meet future development needs, aligned with our development objectives.	Guo et al.(2020)
3b. We formulate learning plans to fill the current knowledge gaps, aligned with our development objectives.	
3c. We formulate learning plans for resource integration and allocation based on our innovation strategy.	
3d. We establish long-term development goals based on our own business situation	
<i>Construct 4: emergent strategic knowledge integration (EKI)</i>	Archer-Brown and Kietzmann (2018)
4a. We promptly integrate newly acquired knowledge with the existing knowledge structure to meet our needs in response to market changes	Guo et al.(2020)
4b. We integrate new feedback from customers or the market to update existing goals or establish new ones	
4c. We integrate new technological information to update existing goals or establish new ones	
4d. We are able to apply new knowledge and technologies to solve challenges in our production and management	
<i>Construct 5: environmental dynamics (ED)</i>	Schilke(2014)
5a. The quantity of products and services we need to deliver is constantly changing	
5b. Our customers regularly request new product ideas and services from our research and marketing departments	
5c. The environment in our market undergoes significant changes	
5d. Consumer demands within our industry are continuously evolving	
<i>Construct 6: firm innovation performance (FIP)</i>	Li et al. (2008)
6a. Our speed of innovative product development exceeds that of our competitors	Forés and Camisón (2016),
6b. Our innovative products are more market-recognized compared to our competitors	Cheng and Shiu (2015)
6c. We have a greater capability than our competitors to leverage cutting-edge technology and process in the development of innovative products	
6d. We are more proficient than our competitors in using efficient workflows to execute innovative projects.	
6e. We are more capable than our competitors in introducing new concepts and ideas into the product development.	
6f. We possess a greater ability to develop entirely new products than our competitors	
Source: Elaborated by authors	

About the authors

Xiaoli Li is a professor at the School of Management, Wuhan Textile University, China. She got her PhD degree from Huazhong University of Science and Technology of China. Her main research interests are knowledge management, technology innovation. She has published more than 40 papers in journals as the first author, chaired two National Social Science Foundation projects, received an award for outstanding achievements in social sciences from the government of Hubei Province, China, in 2022, and also received three awards for outstanding achievements in social sciences from the government of Wuhan, China, and published one academic monograph.

Zihan Peng is a master candidate at the School of Management, Wuhan Textile University, China. Her main research interests are knowledge management, technology innovation.

Kun Li is a master candidate at the School of Management, Wuhan Textile University, China. He got his bachelor degree from the school of management at Henan University of Animal Husbandry and Economy. His main research interests are knowledge management, technology innovation. Kun Li is the corresponding author and can be contacted at: 15713719136@163.com

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: permissions@emeraldinsight.com